Geometry, Topology, and γ-Face Invariance Across N in Event Density Dynamics

Allen Proxmire
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Abstract
ED-Arch-18 maps the global structure of the γ-faces in Event Density (ED) by systematically probing how geometry, topology, and system size shape the three dynamical regimes defined by $\gamma \in \{+1, 0, -1\}$. Across six coordinated sub-experiments (18A–18F), we test γ-face stability under geometry perturbations, topology-conditioned geometry swaps, cross-γ geometry transport, cross-N geometry transport, and manifold stitching along the geometry line connecting N = 4 and N = 6 native slices. The results reveal a striking architectural separation: γ = +1 and γ = −1 are rigid global law-faces, invariant under geometry, topology, and N; their χ□ bands shift only through geometric scaling. In contrast, γ = 0 forms a stitched, N-dependent saddle manifold whose stability depends on geometry scale and whose DECAY→COLLAPSE transition is exposed by cross-geometry and cross-N transport. PREMERGED configurations identify feasibility boundaries rather than dynamical outcomes, showing that geometry constrains which worlds can exist, while γ determines how they evolve. Together, these results demonstrate that the γ-faces are not local artifacts but global dynamical surfaces in the joint space (ψ_2,$d_{\min}$,N,topology), confirming ED's emerging identity as a theory of possibilities: a framework that characterizes not just the behavior of a single world, but the structure of all worlds permitted by the architecture.

1. Introduction

ED-Arch-18 investigates one of the central architectural questions in Event Density:
How do geometry, topology, and dynamical law interact to determine the space of possible ED worlds?

Earlier ED-Arch work established that the three γ-channels —

γ = +1 (inward bias),
γ = 0 (neutral/tangential),
γ = −1 (outward bias) —

produce three distinct dynamical regimes: clean-collapse, orbit/decay, and anomalous-collapse.
But these results were obtained under native geometries at fixed N.
What remained unknown was whether these γ-faces were:

- intrinsic law surfaces,
- geometry-conditioned artifacts,
- topology-dependent,
- N-dependent,
- or some mixture of the above.

ED-Arch-18 resolves this by systematically probing γ-faces under:

- Geometry perturbations (18B)
- Topology-conditioned geometry swaps (18C)

- Cross-γ geometry transport (18D)
- Cross-N geometry transport (18E)
- Manifold stitching along the geometry line (18F)

Across more than 200 controlled simulations, ED-Arch-18 reveals a remarkably coherent structure:

- γ = +1 and γ = −1 are global, rigid law-faces
- γ = 0 is a stitched, N-dependent saddle manifold
- Geometry defines feasibility, not dynamics
- Topology stratifies χ within each face
- N controls the rigidity of the γ-faces

These results show that the γ-faces are not local artifacts but global objects in the ED possibility space, confirming ED's trajectory toward a theory of possibilities.

2. Methods

2.1 Dynamical framework

All simulations use the standard ED update rule:

- Particles move under a γ-biased gradient field
- Collisions are resolved via the ED merge rule
- χ is measured as the normalized time-to-collapse or time-to-decay

Periodic boundary conditions (PBCs) are used throughout

The three γ-channels define the dynamical law:

- γ = +1: inward gradient → clean-collapse
- γ = 0: tangential gradient → orbit/decay
- γ = −1: outward gradient → anomalous-collapse

These channels are the "faces" whose invariance is tested.

2.2 Geometry parameters

Each initial condition is defined by two geometric quantities:

- ψ_2: angular spread (shape descriptor)
- $d_{_min}0$: minimum pairwise distance (scale descriptor)

A geometry-matching operator adjusts particle positions to match a target (ψ_2, d_min0) pair using a binary search over perturbation magnitude.

If the search collapses a pair below MERGE_THR = 23.5 px, the configuration is labeled PREMERGED — a geometric infeasibility, not a dynamical outcome.

2.3 Topologies

Two topologies are used:

- RING: symmetric, periodic, low-density
- CSAT: asymmetric, high-density, encounter-focusing

These topologies define the shape of the initial condition but not the γ-face identity.

2.4 N values

ED-Arch-18 studies two system sizes:

- N = 4: low-N regime, strong geometric constraints
- N = 6: intermediate-N regime, increased rigidity

These two Ns allow us to probe how γ-faces scale and whether they form global manifolds across N.

2.5 Seeds and reproducibility

Each configuration is run with seeds {0, 1, 2}.
Seed-dependence is interpreted structurally:

- Consistent outcomes across seeds → stable region of a γ-face
- Mixed outcomes → phase boundary
- PREMERGED → geometric infeasibility

3. ED-Arch-18A — Baseline γ-Faces at N = 4 and N = 6

ED-Arch-18A establishes the native γ-faces for each topology at each N.
This provides the reference geometry slices used in all subsequent tests.

3.1 Native geometry slices

For each (N, topology), the solver identifies a unique geometry slice:

- RING (N = 4): $\psi_2 = 0.1716$, d_min0 ≈ 63.2
- RING (N = 6): $\psi_2 = 0.2774$, d_min0 = 45.0
- CSAT (N = 4): $\psi_2 = 0.3568$, d_min0 = 34.64
- CSAT (N = 6): $\psi_2 = 0.4307$, d_min0 = 26.5

These slices are topology-conditioned but γ-independent — a key architectural fact.

3.2 γ-face outcomes

Across all seeds and both topologies:

- γ = +1 → clean-collapse
- χ□ bands: 0.05–0.30
- γ = 0 → orbit/decay
- χ□ bands: 4.4–5.0
- γ = −1 → anomalous-collapse
- χ□ bands: 1.4–2.0

These three regimes are cleanly separated and seed-stable.

3.3 Topology-conditioned χ□ ordering

Within each γ-face:

- CSAT collapses faster than RING for $\gamma = +1$
- CSAT decays at a higher plateau for $\gamma = 0$
- CSAT collapses slower than RING for $\gamma = -1$

Topology does not change the outcome, only the intensity.

3.4 Architectural invariant

γ is a dynamical law parameter.
It does not depend on geometry or topology.
All γ-faces live on the same geometry slice for each topology.

This separation — geometry vs topology vs law — is the foundation for the rest of ED-Arch-18.

4. ED-Arch-18B — Geometry-Perturbation Robustness (±10%)

4.1 Purpose

ED-Arch-18B probes whether the γ-faces identified in 18A remain stable when the underlying geometry is perturbed.

For each topology (RING, CSAT) and each $\gamma \in \{+1, 0, -1\}$, we apply:

- $\psi_2 \to 1.10\ \psi_2$, $d_min0 \to 0.90\ d_min0$ (compression)
- $\psi_2 \to 0.90\ \psi_2$, $d_min0 \to 1.10\ d_min0$ (expansion)

and run seeds {0,1,2}.

This tests:

- seed-stability
- channel-stability
- χ□-band stability
- topology ordering
- face separation under metric perturbation

4.2 Results at N = 4

At N = 4, the γ-faces respond asymmetrically to geometric perturbation:

- $\gamma = +1$ remains rigid: clean-collapse across all seeds and perturbations.
- $\gamma = -1$ remains collapse-dominated, but χ□ splits by topology:

- RING: $\chi_{\square} \approx 1.45$
- CSAT: $\chi_{\square} \approx 2.0$ (topology-conditioned collapse channel)

$\gamma = 0$ forms a saddle:

- small perturbations preserve orbit/decay
- larger perturbations push the system into anomalous-collapse
- seed-dependence appears near the saddle boundary

This is the first appearance of the $\gamma = 0$ saddle structure.

4.3 Results at N = 6
At N = 6, the picture changes dramatically:

- $\gamma = +1$: clean-collapse, $\chi_{\square} \in [0.05, 0.30]$
- $\gamma = 0$: orbit/decay, $\chi_{\square} \approx 5.0$
- $\gamma = -1$: anomalous-collapse, $\chi_{\square} \in [1.4, 1.8]$

No perturbation flips a γ-face.
No seed-dependence appears except in PREMERGED cases (geometric infeasibility).
$\chi_{\square}$ bands remain sharply separated.

This is full metric rigidity.

4.4 Architectural invariant
At N = 6, the γ-faces are metric-invariant surfaces.
At N = 4, $\gamma = 0$ is a saddle.

This establishes the first N-dependent structural transition in ED.

5. ED-Arch-18C — Topology-Conditioned Geometry Swaps

5.1 Purpose
18C tests whether γ-faces survive when a topology is forced to use the other topology's geometry.
This isolates the role of topology in shaping the γ-faces.

5.2 N = 6 Results
5.2.1 RING ← CSAT geometry
All configurations PREMERGED.

CSAT geometry is too tight for a 6-ring to realize at $\psi_2 = 0.4307$.

This is a hard feasibility boundary, not a dynamical outcome.

5.2.2 CSAT ← RING geometry
$\gamma = +1 \rightarrow$ clean-collapse

$\gamma = -1 \rightarrow$ anomalous-collapse

$\gamma = 0 \rightarrow$ anomalous-collapse ($\chi\square \approx 2.1$)

This is the first mechanism flip at N = 6.

The tangential channel becomes collapse-susceptible under foreign geometry.

5.2.3 Architectural meaning
$\gamma = +1$ and $\gamma = -1$ are geometry-rigid.

$\gamma = 0$ is geometry-sensitive.

Topology determines feasibility, not γ-face identity.

5.3 N = 4 Results
The same pattern appears, but softer:

RING ← CSAT geometry PREMERGED for seeds 0–1; seed 2 falls back to native geometry.

CSAT ← RING geometry runs cleanly; $\gamma = 0$ flips to anomalous-collapse at large d_min0.

This shows that N = 4 has a softer feasibility boundary, but the same architectural structure.

5.4 Architectural invariant
$\gamma = +1$ and $\gamma = -1$ are topology-invariant and geometry-rigid.
$\gamma = 0$ is geometry-sensitive at both N = 4 and N = 6.

This is the first evidence that $\gamma = 0$ is not a rigid face but a transitional manifold.

6. ED-Arch-18D — Cross-γ Geometry Transport

6.1 Purpose
18D asks a deeper question:

Does each γ-face require its own geometry, or are the γ-faces intrinsic law surfaces?

To test this, we run $\gamma_target \in \{+1, 0, -1\}$ using the geometry slice associated with $\gamma_source \in \{+1, 0, -1\}$.

6.2 Results

6.2.1 $\gamma = +1$
Always clean-collapse

$\chi\square$ shifts monotonically with d_min0

Completely insensitive to geometry source

6.2.2 γ = −1
Always anomalous-collapse

χ□ shifts monotonically with $d_{min}0$

Completely insensitive to geometry source

6.2.3 γ = 0
Native geometry → orbit/decay

Foreign geometry → anomalous-collapse

χ□ ≈ 2–3

Collapse mechanism: PBC corner-wrap collision

This is the same gate-failure mechanism seen in 18C.

6.3 Architectural invariant
γ = +1 and γ = −1 are intrinsic law surfaces.
γ = 0 is a geometry-conditioned transitional manifold.

This is the first time ED shows a clean separation between law and geometry.

7. ED-Arch-18E — Cross-N γ-Face Transport

7.1 Purpose
18E asks whether γ-faces are N-local artifacts or global surfaces in the joint space(ψ_2,$d_{min}0$,N,topology).

The test transports geometry slices between N = 4 and N = 6, holding topology fixed, and observes whether γ-faces retain their identity.

This isolates the role of system size in shaping the γ-faces.

7.2 Results

7.2.1 γ = +1 and γ = −1 are globally rigid across N
Across all cross-N transports:

γ = +1 → clean-collapse

γ = −1 → anomalous-collapse

χ□ shifts monotonically with d_min0, exactly as a geometric scaling law predicts

No mechanism flips

No seed-dependence (except PREMERGED feasibility failures)

This is the strongest evidence in Arch-18 that $\gamma = +1$ and $\gamma = -1$ are true law-faces, not geometric or topological constructs.

7.2.2 $\gamma = 0$ collapses whenever d_{min} is upshifted across N
Whenever geometry from a larger-N slice (larger radius) is injected into a smaller-N target, or vice-versa:

$\gamma = 0 \rightarrow$ COLLAPSE

$\chi \approx 2–4$

Mechanism: PBC corner-wrap collision

Channel identity: orbit → anomalous-collapse

This is the same gate-failure mechanism seen in 18C and 18D, but now universal across N.

7.2.3 PREMERGED patterns reveal N-dependent feasibility boundaries
N6 geometry → N4 targets often PREMERGE (too tight)

N4 geometry → N6 targets never PREMERGE (too loose)

CSAT seed-0 PREMERGE is a stable structural feature

RING never PREMERGES under N4→N6 transport

This yields the first cross-N feasibility map:

The feasible geometry region expands with N.
The infeasible region grows when compressing geometry downward across N.

7.3 Architectural invariant
$\gamma = +1$ and $\gamma = -1$ are global law-faces across N.
$\gamma = 0$ is an N-dependent saddle manifold whose stability depends on geometry scale.

This is the first time ED shows a clean separation between law, geometry, and system size.

8. ED-Arch-18F — Manifold Stitching Along the Geometry Line

8.1 Purpose
18F performs the final test of Arch-18:

Does the $\gamma = 0$ face form a continuous manifold across N and topology?

To answer this, we trace $\gamma = 0$ along the geometry line connecting the N = 4 and N = 6 native slices, sampling:

- N4 geometry
- midpoint geometry
- N6 geometry

for both RING and CSAT, at both N = 4 and N = 6.

This reveals the shape of the $\gamma = 0$ manifold.

8.2 Results

8.2.1 N = 6 (RING and CSAT)
A clean, monotone transition:

- Native slice → DECAY (χ ≈ 4.4–5.0)
- MID slice → fast DECAY (χ ≈ 2.0–2.6)
- N4 slice → COLLAPSE (χ ≈ 2.0–2.7)

All seeds agree (except PREMERGED feasibility cases).
This is a smooth DECAY → COLLAPSE transition along the geometry line.

8.2.2 N = 4 (RING and CSAT)
The same transition appears, but earlier and sharper:

- Native slice → DECAY
- MID slice → COLLAPSE (all seeds for RING; mixed for CSAT)
- N6 slice → mostly PREMERGED

This shows that N = 4 intersects the $\gamma = 0$ manifold past the transition, while N = 6 intersects it through the transition.

8.2.3 Seed-dependence at the phase boundary
At N = 4 CSAT MID/N6 slices:

- one seed → DECAY
- one seed → COLLAPSE
- one seed → PREMERGED

This is not noise — it is the signature of a phase boundary.

8.3 Architectural invariant
$\gamma = 0$ is a continuous, stitched manifold across N and topology, with a well-defined DECAY → COLLAPSE transition as d_min increases.

N = 4 and N = 6 are simply different cuts through the same surface.

9. Global Architectural Synthesis

ED-Arch-18 reveals the global structure of the γ-faces in ED.

9.1 The three γ-faces as global objects

Across all tests:

- $\gamma = +1 \rightarrow$ clean-collapse
- $\gamma = 0 \rightarrow$ orbit/decay or anomalous-collapse depending on geometry scale
- $\gamma = -1 \rightarrow$ anomalous-collapse

These are not local behaviors — they are global dynamical surfaces.

9.2 Geometry–topology–law separation

Arch-18 shows a clean three-way separation:

- Geometry defines feasibility (PREMERGED regions)
- Topology stratifies χ within each face
- γ defines the dynamical regime

This is the hallmark of a generative ontology.

9.3 N-scaling

At $N = 4$, $\gamma = 0$ is a saddle

At $N = 6$, $\gamma = 0$ becomes semi-rigid

$\gamma = \pm 1$ remain rigid across N

This suggests that as N increases, the γ-faces approach full rigidity.

9.4 ED as a theory of possibilities

The γ-faces define the allowed dynamical regimes.
Geometry defines which worlds are feasible.
Topology defines how strongly each regime expresses itself.
N defines how rigid the regimes become.

This is exactly the structure expected of a theory of possibilities.

10. Conclusion

ED-Arch-18 provides the most comprehensive structural map of the γ-faces to date.
Across geometry perturbations, topology swaps, γ-transport, N-transport, and manifold stitching, the architecture is consistent and law-like:

- $\gamma = +1$ and $\gamma = -1$ are rigid global faces
- $\gamma = 0$ is a stitched, N-dependent saddle manifold

- Geometry defines feasibility, not dynamics
- Topology modulates χ but never flips outcomes
- N controls the rigidity of the γ-faces

Together, these results show that the γ-faces are not artifacts of specific geometries or topologies, but global architectural surfaces in the ED possibility space.

ED-Arch-18 therefore marks a major step in establishing ED as a theory of possibilities — a framework that describes not just what happens in one world, but what kinds of worlds can exist at all.